The Climate Confidence Gap: A Claim-Source Audit Method

Abstract

The IPCC states its confidence finding by finding, and a public claim that drops that confidence, and so changes what is claimed, misreports the science, in either direction. Six findings show what the assessments actually say where public debate is loudest. Climate sensitivity is still a range: 1.5 to 4.5 degrees Celsius in 1979 and again in 2013, and 2.5 to 4.0 since 2021. The high-emissions scenario often called business as usual is one the Sixth Assessment Report judges of low likelihood. Models overstated upper tropical tropospheric warming from 1979 to 2014 by at least 0.1 degrees per decade, with medium confidence. Hot extremes are virtually certain to have increased, while human influence on regional meteorological drought trends has generally low confidence. Scientists inside the field flagged two of them in Nature. Overstated confidence costs the science its credibility when readers find the gap, and the remedy costs nothing: quote the report's own confidence terms. The paper gives the procedure for checking any public claim against its source.

1. The Standard We Are Applying

A summary should preserve what its source says and how firmly it says it. The Intergovernmental Panel on Climate Change (IPCC) distinguishes confidence in an assessment (how well evidence and expert agreement support it) from the likelihood of an outcome (its assessed probability); those terms have defined uses. An audit should check whether a public statement changes the finding, its scope or its uncertainty. Simplifying the language can be accurate. The test is whether the simplification changes the claim.

The six topics below are comparisons worth auditing. None of them is a completed claim-source pair yet: someone must first gather dated public quotations alongside the assessment passages they cite, and only then can anyone report how large or how common a communication gap is.

A reproducible claim-source record

An audit is only as good as its record, so the record has to be fixed before any example is chosen. For each comparison, record the public quotation, speaker or outlet, date, stable link and exact proposition asserted. Record the assessment passage with its section or page, publication date, scenario, scope and uncertainty language. An assessment published after the public statement cannot establish that the earlier statement misrepresented it.

Use a coding scheme (a fixed set of categories for classifying each comparison) that permits agreement, stronger public certainty, weaker public certainty, changed scope, missing conditions and an unassessable match. Two coders should judge independently, preserve disagreements and document their resolution. A prevalence estimate (a figure for how common such gaps are) additionally requires an outlet set, date window and inclusion rule fixed before selecting examples. Report all eligible matches, including accurate summaries.

2. Equilibrium Climate Sensitivity

Climate sensitivity shows scientific progress alongside remaining uncertainty: the assessed range held still for decades, then narrowed, and it is still a range. In the assessment record, the Charney Report (National Research Council, 1979), chaired by the meteorologist Jule Charney, estimated eventual warming after doubled CO2 at 1.5 to 4.5 degrees Celsius. The IPCC's Fifth Assessment Report (AR5; IPCC, 2013) gave the same likely interval (a range judged at least 66

percent probable) in 2013 (Working Group I, Summary for Policymakers, SPM D.2). The Sixth Assessment Report (AR6; IPCC, 2021) narrowed its likely interval to 2.5 to 4.0 degrees, with a best estimate of 3.0 degrees and high confidence in that likely interval (WGI, SPM A.4.4).

A best estimate is not an exact prediction. The remaining interval also does not negate the separate finding that human influence has warmed the atmosphere, ocean and land (AR6 WGI, SPM A.1). An audit should ask whether a summary confuses those two questions.

3. Scenarios Treated as Forecasts

A scenario asks what would happen under specified emissions. It is not automatically a forecast that those emissions will occur. Zeke Hausfather and Glen Peters (2020) challenged the routine use of RCP8.5, the highest-emissions Representative Concentration Pathway, as a business-as-usual baseline.

The report holds two things together here. In the source, AR6 WGI Section 1.6.1.4 states that the report generally attaches no likelihoods to its scenarios. The same section describes high-emissions pathways such as RCP8.5 and SSP5-8.5, its counterpart among the newer Shared Socioeconomic Pathways, as having low likelihood in light of recent energy-sector developments. This is a narrative assessment of plausibility, not a calibrated numerical probability assigned to the scenario.

An audit should preserve both points. A useful high-emissions calculation need not be the expected future, and the plausibility discussion does not make the scenario's conditional physical consequences irrelevant. Check what the public statement claims, against the source available on its date.

4. The Hot Model Problem

A raw model ensemble (the pooled output of many climate models) and an assessed projection need not give the same range. Hausfather and colleagues (2022) discuss models from the sixth Coupled Model Intercomparison Project (CMIP6) with unusually high climate sensitivity and warn against using their projections without considering other evidence.

The assessment had already made that adjustment. In the source, AR6 combined scenario-based model projections with observations of past warming and its assessed climate sensitivity and transient response, the warming reached when gradually rising CO2 first doubles (WGI Section 4.3.4 and Box 4.1). Its assessed global temperature ranges therefore differ from the unadjusted ensemble range. An audit should identify which quantity a public summary reports before comparing it with AR6.

5. Homogenization of the Surface Record

A station's record can change because the climate changes or because the station moves, replaces an instrument or changes observing practice. Homogenization aims to separate those effects.

The published method checks a station against nearby stations, so each adjustment carries a stated reason that another analyst can test. In the documentation, version 4 of the Global Historical Climatology Network monthly dataset (GHCNm), kept by the National Oceanic and Atmospheric Administration (NOAA), compares a station with nearby correlated records to detect abrupt shifts affecting that station. It then estimates corrections from the differences (Menne and Williams, 2009; NOAA NCEI, GHCNm documentation). The direction of a correction alone shows neither manipulation nor accuracy.

An audit should state the dataset and version, the affected period and the adjustment being discussed. Berkeley Earth's separate averaging method provides another published comparison (Rohde et al., 2013). Agreement between analyses matters, as do shared observations and methodological differences; none should be hidden by a claim about adjustments in general.

6. The Tropical Troposphere

A model-observation difference must be tied to its altitude, region and period. In the assessment's terms, AR6 assesses with medium confidence that CMIP5 and CMIP6 models overestimate warming in the upper tropical troposphere (the lowest layer of the atmosphere) during 1979-2014 by at least 0.1 degrees Celsius per decade, partly because they overestimate the tropical sea-surface warming pattern (WGI Section 3.3.1.2.1).

The report also notes medium confidence that the upper tropical troposphere has warmed faster than the near-surface since at least 2001 (Section 3.3.1.2, referring to Section 2.3.1.2). These statements answer different comparisons. An audit should retain the model discrepancy without turning it into a claim that no upper-air amplification (faster warming aloft than at the surface) has been observed.

7. Extreme Event Attribution

Different extremes support different conclusions, and one sentence of assessment can attach a likelihood term to a trend and a separate confidence term to its cause. In the assessment's own terms, AR6 calls the increase in hot-extreme frequency and intensity across most land regions since the 1950s virtually certain (at least 99 percent probable), with high confidence that human influence is the main driver (WGI, SPM A.3.1).

Drought needs a finer distinction, because the assessment reaches different conclusions for different kinds of drought. In the assessment's own terms, Section 11.6.4.5 reports generally low confidence in human influence on regional meteorological-drought trends (trends in shortfalls of rain and snow), with exceptions. It reports medium confidence in human contributions to increased agricultural and ecological drought (too little soil moisture for crops and ecosystems) in some regions, including the Mediterranean and western North America. It also distinguishes the effects of climate, land use and water management on hydrological drought (low water in rivers, lakes and reservoirs).

The same section separates trends from individual events: an event can be attributed in a region without a detected long-term trend. A summary of drought everywhere cannot stand in for these different findings. An audit must match drought type, location, period and causal question before judging the claim.

8. Why Accurate Communication Matters

When a writer turns a conditional projection into an unconditional prediction without saying so, a reader who later discovers it can lose trust. The audit must also record accurate summaries and appropriate simplifications. Selecting only conspicuous disagreements builds the conclusion into the sample.

9. Recommendation

Use plain language that preserves the source's meaning. Keep a confidence or likelihood qualifier when omitting it would make the claim stronger or weaker. State the scenario and geographical scope when they affect the conclusion. A summary need not reproduce every technical term to be accurate; it needs to preserve the distinctions on which the finding depends. How readers understand

alternative wording is a further empirical question.

Conclusion

Put a public climate statement beside the assessment it cites, and any difference in certainty between them shows. This paper gives the procedure for doing that: quote both, date both, and record the scenario, scope and uncertainty language the source used. The assessment record holds progress and remaining uncertainty together, and an audit that reports one has to report the other. Published comparisons would show which public summaries report their sources accurately and which need correcting.

References

Hausfather, Z., and Peters, G. P. (2020). Emissions: the 'business as usual' story is misleading. *Nature*, 577, 618-620. https://doi.org/10.1038/d41586-020-00177-3

Hausfather, Z., Marvel, K., Schmidt, G. A., Nielsen-Gammon, J. W., and Zelinka, M. (2022). Climate simulations: recognize the 'hot model' problem. *Nature*, 605, 26-29. https://doi.org/10.1038/d41586-022-01192-2

IPCC (2013). *Climate Change 2013: The Physical Science Basis*. Contribution of Working Group I to the Fifth Assessment Report. Summary for Policymakers, Section D.2. https://www.ipcc.ch/report/ar5/wg1/

IPCC (2021). *Climate Change 2021: The Physical Science Basis*. Contribution of Working Group I to the Sixth Assessment Report. The passages used here are SPM A.1, A.3.1 and A.4.4; Section 1.6.1.4; Sections 3.3.1.2 and 3.3.1.2.1; Section 4.3.4 and Box 4.1; and Section 11.6.4.5. https://www.ipcc.ch/report/ar6/wg1/

Menne, M. J., and Williams, C. N., Jr. (2009). Homogenization of temperature series via pairwise comparisons. *Journal of Climate*, 22, 1700-1717. https://doi.org/10.1175/2008JCLI2263.1

National Research Council (1979). *Carbon Dioxide and Climate: A Scientific Assessment* (the Charney Report). National Academies Press. https://doi.org/10.17226/12181

NOAA National Centers for Environmental Information (n.d.). *Global Historical Climatology Network monthly (GHCNm)*. Version 4 documentation, Homogenization section. Accessed 6 October 2026. https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-monthly

Rohde, R., et al. (2013). Berkeley Earth temperature averaging process. *Geoinformatics & Geostatistics: An Overview*, 1(2). https://doi.org/10.4172/gigs.1000103